

Parabolic induction

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G/E conn'd reductive group over $E =$ no local field.

$$E \supset \mathcal{O}_E \ni \pi$$

$\downarrow$

work over Perf_k : perfectoid spaces / k

$$\overline{E}_\pi = k \leftarrow \mathbb{E}_\pi = \mathcal{O}_E / \pi$$

$$- \text{Bun}_G : \text{Perf}_k \rightarrow \text{Cpt}, \quad S \mapsto \text{Bun}(X_S)$$

$$- \text{Div}^1 : \text{Perf}_k \rightarrow \text{Cpt}, \quad S \mapsto \{ \text{deg. 1 Cartier divisor on } X_S \}$$

$$\text{Spd } \check{E} / \varphi^\mathbb{Z} \quad \check{E} | E \quad \begin{array}{l} \text{completion of} \\ \text{max'l unram. ext'n} \end{array}$$

$$|\text{Bun}_G| \simeq B(G) = G(\check{E}) / \varphi\text{-conj.} \simeq G\text{-Isoc}_k \xrightarrow{(\nu, \kappa)} (X_\star^+(T))_{\mathcal{O}_k}^\Gamma \times \pi_1(G\check{E})_\Gamma$$

$\text{Isoc} : \check{E}\text{-vector space with}$
 $\sigma\text{-semi linear automorphism}$

$$K : \pi_0(|\text{Bun}_G|) \xrightarrow{\sim} \pi_1(G)_\Gamma$$

$$\text{Bun}_G^\alpha \hookrightarrow \alpha$$

$\hookrightarrow$ contains a unique open point (called (semi)stable) basic point
 $b \in B(G)$ called basic.

$$\text{Bun}_G = \coprod_{b \in B(G)} \text{Bun}_G^b \quad \text{into locally closed substack.} \quad \text{Bun}_G^b = [* / \text{aut.}]$$

$$b \text{ basic, } \text{Bun}_G^b = [* / G_b(E)], \quad b \text{ general, } \text{Bun}_G^b = \left[[* / G_b(E) \rtimes \begin{array}{l} \text{dim} \langle \text{zp}, \nu_b \rangle \\ \text{unipotent group} \\ \text{diamond} \end{array} \right]$$

$$G_b(R) = \{ g \in G(R \otimes_E \check{E}) : gb = b \sigma(g) \}$$

$$D(\text{Bun}_G^b) \simeq D(* / G_b(E)) = D(\text{Rep}^{\text{sm}}(G_b(E))) \quad \text{for all } b$$

$b \rightsquigarrow \xi_b$ corresponds to Bun_G^b

b basic, $\text{Bun}_G \simeq \text{Bun}_{G,b}$, $\xi \mapsto \underline{\text{Isom}}_G(\xi, \xi_b)$

I finite set

$$\begin{array}{c} \text{Hecke}_G^I \\ \swarrow \downarrow \searrow \\ \xi_1 \quad \downarrow p_1 \quad \downarrow p_2 \quad \downarrow \\ \text{Bun}_G \quad (\xi_2, D) \text{Bun}_G \times (D|_{X_S})^I \end{array} \quad \begin{array}{l} : S \mapsto \left(\begin{array}{l} D = \bigcup_i D_i \subset X_S \\ D_i \in (D|_{X_S})^I(S), \quad \xi_1, \xi_2 \in \text{Bun}_G(X_S), \\ \alpha: \xi_1|_{X_S \setminus D} \simeq \xi_2|_{X_S \setminus D} \end{array} \right) \\ \uparrow \\ \text{isomorphic along } D \end{array}$$

$\xi_1, \xi_2 \in \text{Bun}_G$ (normal nbhd of $D \subset X_S$)

Geometric Satake

$$\text{Rep}_{\hat{G}^I}(\text{Rep}^{\text{cont}}(W_E^I)) \hookrightarrow D(\text{Hecke}_G^I)$$

$$\begin{array}{ccc} V & \searrow & \downarrow \\ & S_V & D(\text{Hecke}_G^I) \end{array}$$

eg. G split, $\text{Rep}(\hat{G}) \rightarrow D(\text{Hecke}_G)$, $\text{Hecke} = \bigsqcup_{\mu \in X_+^*(G)} \text{Hecke}_\mu$

$$\text{Hecke} = \bigsqcup_{\mu \in X_+^*(G)} \text{Hecke}_\mu$$

$V_\lambda =$ highest wt

rep of λ

$\mapsto$

$$\Lambda_{\text{Hecke}_\mu}[\dim]$$

}

middle ext'n

}

pull back to Hecke

$$S_V|_{\text{Hecke}_\mu} = \Lambda[\text{shift}]$$

$$V \rightsquigarrow S_V \rightsquigarrow T_V = p_2! \left(p_L^*(-) \otimes S_V \right) : D(\text{Bun}_G) \longrightarrow D(\text{Bun}_G \times (\text{Div}^1)^I)$$

$$\searrow$$

$$D(\text{Bun}_G)^{S/BW_E^I}$$

$$\text{Rep}((A \rtimes Q)^I) \longrightarrow \text{End}(D(\text{Bun}_A))^{W_E^I} \quad \text{functorial in } I$$

$$D(\text{Div}^1) = D(* / w_E)$$

$$Div^1 = Spd \check{E} / \varphi$$

$$= Spd C / I_E \propto \psi$$

$$h \text{ split} \quad \text{Rep}(\hat{h}^{\mathbf{Z}}) \longrightarrow \text{End}(D(Bun_h))$$

$$\begin{array}{ccc} \{1,2\} & & \hat{L}^2 \\ \downarrow & & \uparrow \\ |1\rangle & & \hat{L}_z \end{array}$$

$$\downarrow$$

$$\text{Rep}(\hat{a})$$

$$\rightsquigarrow \text{Exc}(W_E, \hat{G}) \longrightarrow \text{End}(\text{id}_{D(B_{\text{un}}^{\text{un}})})$$

$$\mathbb{Q} \vdash \pi_1(\hat{G})_{\text{tor}} \quad \text{''}$$

$$\mathcal{O}\left(\frac{Z^1(W_E, \hat{G})}{\hat{G}}\right) = \text{Colim}_{(n, F_n \rightarrow W_E)} \mathcal{O}(Z^1(F_n, \hat{G}))^{\hat{G}}$$

$$\lim_{n \cdot F_n \rightarrow W_E} \left(\hat{G} \otimes Q \right)^n / \hat{G}$$

$$\hat{G} \text{ split. } \mathcal{O}(\hat{G}^n / \hat{G}) \longrightarrow \text{End}(\text{id}_{D(\text{Bun}_G)}) = \mathbb{Z}^{\text{geom}}(G)$$

$\left\{ \begin{array}{l} \text{excursion datum: } I, V, \beta, (r_i)_{i \in I}: \\ \quad I \text{ finite set}, \quad \alpha: 1 \rightarrow V|_A \\ \quad V \in \operatorname{Rep}(\hat{\Lambda}^I), \quad \beta: V|_A \rightarrow 1 \\ \quad D = \{(r_i)_i \in W_E^I\} \end{array} \right.$

$$S_D : id = T_1 \xrightarrow{T_\alpha} T_V \xrightarrow{(v_i)_i} T_V \xrightarrow{T_\beta} T_2 = id \quad \text{excursion operator}$$

$$t_D \in \mathcal{O}(\hat{a} \backslash \hat{a}^I / \hat{a}) \quad g_i \xrightarrow{t_D} \langle \beta, g_i \alpha \rangle$$

$$I = \{0, 1, \dots, n\} \quad \begin{matrix} \nearrow (1, g_1, \dots, g_n) \quad S_1 \\ \downarrow (g_1, \dots, g_n) \end{matrix}$$

$$\mathcal{O}(\hat{a}^n / \hat{a})$$

S_D depends only on t_D

/ Bernstein centre

$$\mathbb{I}_{\hat{a}}: \text{Exc}(\hat{G}, w_E) \longrightarrow \mathbb{Z}^{\text{geom}}(\hat{a}) \longrightarrow \mathbb{Z}(\hat{a}(E))$$

$$\mathbb{I}_{\hat{a}}^b: \text{Exc}(\hat{G}, w_E) \longrightarrow \mathbb{Z}(\hat{G}_b(E))$$

$$\begin{matrix} \nearrow \mathbb{I}_{\hat{G}_b} \\ \text{Exc}(\hat{G}_b, w_E) \end{matrix}$$

$$\text{Th.} \quad \text{Exc}(\hat{G}, w) \xrightarrow{\mathbb{I}_{\hat{a}}^b} \mathbb{Z}(\hat{G}_b(E))$$

$$\begin{matrix} \downarrow \\ \text{Exc}(\hat{G}_b, w) \end{matrix} \quad \nearrow \mathbb{I}_{\hat{G}_b}$$

commutes.

$$\hat{G}_b \rtimes w_E \longrightarrow \hat{G} \rtimes w_E$$

$$\text{split case} \quad \hat{G}_b \longrightarrow \hat{G} \quad \text{Levi}$$

Subtle twist from geometric Satake.

centralizer of

$$V_b \in X_{\text{st}}^+(T)_{\text{cl}}$$

$$\mathbb{Z}^1(\hat{G}_b) \longrightarrow \mathbb{Z}^1(\hat{G})$$

$$\Gamma_{\hat{G}} \in \Lambda$$

$$\varphi \longmapsto \left[w \mapsto (2P_{\hat{G}} - 2P_{\hat{G}_b})(\sqrt{q})^{|w|} \varphi(w) \right]$$

Cor. $p \in G$ parabolic, $p \rightarrow M$ Levi quotient. then for all $\sigma \in \text{Rep}(M(E))$,

$$\text{Exc}(\hat{G}, w) \xrightarrow{\Phi_G} \text{End}(\text{Ind}_{P(E)}^{G(E)} \sigma)$$

$$\begin{array}{ccc} \downarrow & \searrow \Phi_G^b & \uparrow \\ \text{Exc}(\hat{M}, w) & \xrightarrow{\Phi_M} & \text{End}(\sigma) \end{array}$$

commutes.

Proof of Cor. $\mu: G_m \rightarrow G$ cochar w dynamic parabolic P , $b = \mu(\pi^{-1}) \in B(G)$

$$M = G_b$$

Remains to show

$$\begin{array}{ccc} \text{Exc}(\hat{G}, w) & \longrightarrow & \text{End}(\text{Ind}_{P(E)}^{G(E)} \sigma) = \text{End}\left(T_{\mu^{-1}}(A) \Big|_{\text{Bun}_G^1}\right) \\ & \searrow & \uparrow \\ & & \text{End}(\sigma) \end{array}$$

$$\sigma \rightsquigarrow A \in D(\text{Bun}_G^b) = D(* / M(E))$$

Claim 1. $T_{\mu^{-1}}(A) \Big|_{\text{Bun}_G^1} \in D(* / G(E))$ given by $\text{Ind}_{P(E)}^{G(E)} \sigma[-d](-\frac{d}{2})$

$\Rightarrow$ by Excursion operators commute with Hecke operators

$$D = (I, \alpha, \beta, V, (\gamma_i)_i)$$

$$\left[\begin{array}{l} A \rightarrow T_V(A) \xrightarrow{\cdot(\gamma_i)} T_V(A) \rightarrow A \\ T_{\mu^{-1}}(A) \rightarrow T_{\mu^{-1}}(T_V(A)) \rightarrow T_{\mu^{-1}}(T_V(A)) \rightarrow T_{\mu^{-1}}(A) \end{array} \right]$$

$$\left[T_{\mu^{-1}}(A) \rightarrow T_V(T_{\mu^{-1}}(A)) \rightarrow T_V(T_{\mu^{-1}}(A)) \rightarrow T_{\mu^{-1}}(A) \right]$$

How to compute $T_{\mu^{-1}}(A) \Big|_{\text{Bun}_G^1}$?

$$\begin{array}{ccc} & \text{Hecke}_G^1 \Big|_{\text{Bun}_G^b \times \text{Bun}_G^1} & \\ \swarrow & & \searrow \\ \text{Bun}_G^b & & \text{Bun}_G^1 \subset \text{Bun}_G \end{array}$$

Analyze: $\mu = \{ \xi_1, \xi_2, \xi_1 \xrightarrow{\quad} \xi_2 \}$ modification with type bounded by μ

$\downarrow$ $\downarrow$
 trivial ξ_2
 G -torsion

$\subset \text{Hecke}_\mu$

Want to know $S_{V_\mu} \in D(\text{Hecke} \mid \text{Bun}_G^b \times \text{Bun}_G^1)$

$$S_{V_\mu} \Big|_\mu \quad \mu \subset \text{Hecke}_\mu$$

||

$$\Lambda[\text{shift}] \quad G(E) \backslash G(E) \times \text{Bun}_G^b \quad b = \mu(\pi^{-1})$$

Univ. $\mu \simeq G(E) \backslash G(E) / P(E) = * / P(E)$

Standard modification

$$0 \rightarrow \mathcal{O}(1) \quad G_m\text{-torsion}$$

$$* / P(E) \quad \Lambda[-d](-\frac{d}{2})$$

pushout along $G_m \xrightarrow{\mu} G$

$$P_1 \swarrow \quad \searrow P_2$$

$$* / M(E) \quad * / G(E)$$

$$T_{\mu^{-1}}(A) \Big|_{\text{Bun}_G^b} = P_2! \left(P_1^*(-) \otimes \Lambda[-d](-\frac{d}{2}) \right)$$

$$= \text{Ind}_{P(E)}^{G(E)} (-) \cdot (\text{shift})$$

Similarly, $T_\mu(B) \Big|_{\text{Bun}_G^b} \quad B \simeq \tau \in \text{Rep}(G(E))$

$\rightsquigarrow$ compatibility with parabolic restriction